PYTHON PROGRAMMING LABORATORY

14 Dashboard-Based Real-World Case Studies

Based on the prescribed syllabus: Python Basics, Strings, Lists, Functions & Modules, Regular Expressions, File Handling, Database Connectivity, Transactions and Error Handling.

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Experiment 1: University Student Academic Dashboard

Syllabus Topic: Variables, Data Types, Input/Output and Operators

Question / Problem Statement

Write a Python program to read a student's ID, name, department and marks in five subjects. Calculate total, average and percentage and display a University Student Academic Dashboard.

Requirements

- Write and execute the Python program.
- Display the requested information as a clear dashboard.
- Test the program with different input values.

Algorithm

1. Read/prepare the required data.

2. Process the data using the prescribed Python concept.

3. Calculate the required results.

4. Display the dashboard and verify the result.

Python Program

```
sid=input("Student ID: "); name=input("Name: "); dept=input("Department: ")
marks=[]
```

```
for subject in ["Python","DBMS","Mathematics","OS","Networks"]:
    marks.append(float(input(subject+" marks: ")))
total=sum(marks); average=total/5; percentage=total/500*100
print("\n========================================")
print("       UNIVERSITY STUDENT DASHBOARD")
print("========================================")
print("Student ID :",sid); print("Name       :",name); print("Department :",dept)
print("Total      :",total); print("Average    :",round(average,2))
print("Percentage :",round(percentage,2),"%")
print("========================================")
```

Sample Dashboard Output

```
========================================
       UNIVERSITY STUDENT DASHBOARD
========================================
Student ID : 2026001
Name       : Ayesha
Department : CSE
Total      : 408
Average    : 81.6
Percentage : 81.6 %
========================================
```

Result

The program was executed successfully and the required dashboard was generated.

Viva / Practice Questions

1. What are variables and data types?

2. Which operators are used for percentage calculation?

3. Why does input() initially return a string?

Experiment 2: Banking Account Dashboard

Syllabus Topic: if, if-else and nested if

Question / Problem Statement

Write a Python program for a banking dashboard. Accept account details and perform deposit, withdrawal or balance enquiry using conditional statements. Do not permit withdrawal beyond the available balance.

Requirements

- Write and execute the Python program.
- Display the requested information as a clear dashboard.
- Test the program with different input values.

Algorithm

1. Read/prepare the required data.

2. Process the data using the prescribed Python concept.

3. Calculate the required results.

4. Display the dashboard and verify the result.

Python Program

```
account=input("Account Number: "); customer=input("Customer Name: ")
balance=float(input("Opening Balance: ₹"))
print("1.Deposit  2.Withdraw  3.Balance")
choice=input("Choice: ")
if choice=="1":
    amount=float(input("Deposit: ₹"))
    if amount>0: balance+=amount; msg="Deposit Successful"
    else: msg="Invalid Amount"
elif choice=="2":
    amount=float(input("Withdrawal: ₹"))
    if amount<=0: msg="Invalid Amount"
    elif amount<=balance: balance-=amount; msg="Withdrawal Successful"
    else: msg="Insufficient Balance"
elif choice=="3": msg="Balance Enquiry Successful"
else: msg="Invalid Choice"
print("\n================================")
print("         BANKING DASHBOARD")
print("================================")
print("Account :",account); print("Customer:",customer)
print("Status  :",msg); print("Balance : ₹",round(balance,2))
print("================================")
```

Sample Dashboard Output

```
================================
         BANKING DASHBOARD
================================
Account : 10024567
Customer: Ahmed
Status  : Deposit Successful
Balance : ₹ 55250.0
================================
```

Result

The program was executed successfully and the required dashboard was generated.

Viva / Practice Questions

1. How does if-elif-else work?

2. Why is balance checked before withdrawal?

3. Give one example of nested if.

Experiment 3: Hospital & Diagnostic Centre Dashboard

Syllabus Topic: Conditional Statements

Question / Problem Statement

Write a Python program to calculate a hospital patient's bill. Accept room type, days, doctor charges, diagnostic charges and medicine charges. Apply a discount based on the total bill and display the final dashboard.

Requirements

- Write and execute the Python program.
- Display the requested information as a clear dashboard.
- Test the program with different input values.

Algorithm

1. Read/prepare the required data.

2. Process the data using the prescribed Python concept.

3. Calculate the required results.

4. Display the dashboard and verify the result.

Python Program

```
pid=input("Patient ID: "); name=input("Patient Name: ")
room=input("Room (General/Private): "); days=int(input("Days: "))
doctor=float(input("Doctor Charges: ₹")); diagnostic=float(input("Diagnostic: ₹"))
medicine=float(input("Medicine: ₹"))
if room.lower()=="general": room_charge=days*2000
elif room.lower()=="private": room_charge=days*4000
else: room_charge=days*2500
total=room_charge+doctor+diagnostic+medicine
if total>=20000: discount=total*.10
elif total>=10000: discount=total*.05
else: discount=0
print("\n========================================")
print("       HOSPITAL & DIAGNOSTIC DASHBOARD")
print("========================================")
print("Patient:",pid,name); print("Room Charges: ₹",room_charge)
print("Total Bill : ₹",round(total,2)); print("Discount   : ₹",round(discount,2))
print("Net Amount : ₹",round(total-discount,2))
print("========================================")
```

Sample Dashboard Output

```
========================================
       HOSPITAL & DIAGNOSTIC DASHBOARD
========================================
Patient: P1025 Ayesha
Room Charges: ₹ 8000
Total Bill : ₹ 14000
Discount   : ₹ 700
Net Amount : ₹ 13300
========================================
```

Result

The program was executed successfully and the required dashboard was generated.

Viva / Practice Questions

1. How is elif used?

2. How is the discount selected?

3. How could you add different diagnostic tests?

Experiment 4: Supermarket & Shopping Mall Dashboard

Syllabus Topic: for, while, break and continue

Question / Problem Statement

Write a Python program to enter multiple supermarket products, quantities and prices. Use a loop, stop entry when the user types done, skip invalid quantities, calculate subtotal and discount, and display a sales dashboard.

Requirements

- Write and execute the Python program.
- Display the requested information as a clear dashboard.
- Test the program with different input values.

Algorithm

1. Read/prepare the required data.

2. Process the data using the prescribed Python concept.

3. Calculate the required results.

4. Display the dashboard and verify the result.

Python Program

```
subtotal=0; items=0
while True:
    product=input("Product (done to finish): ")
    if product.lower()=="done": break
    qty=int(input("Quantity: "))
    if qty<=0:
        print("Invalid quantity"); continue
    price=float(input("Price: ₹")); subtotal += qty*price; items += qty
if subtotal>=5000: discount=subtotal*.10
elif subtotal>=2500: discount=subtotal*.05
else: discount=0
print("\n========================================")
print("     SUPERMARKET & SHOPPING MALL")
print("             SALES DASHBOARD")
print("========================================")
print("Items:",items); print("Subtotal: ₹",subtotal)
print("Discount: ₹",discount); print("Final: ₹",subtotal-discount)
print("========================================")
```

Sample Dashboard Output

```
========================================
     SUPERMARKET & SHOPPING MALL
             SALES DASHBOARD
========================================
Items: 10
Subtotal: ₹ 3200
Discount: ₹ 160
Final: ₹ 3040
========================================
```

Result

The program was executed successfully and the required dashboard was generated.

Viva / Practice Questions

1. Difference between for and while?

2. What does break do?

3. What does continue do?

Experiment 5: University Student Feedback / String Dashboard

Syllabus Topic: Strings: accessing, operations, slicing, functions and methods

Question / Problem Statement

Write a Python program to analyze student feedback. Display character count, word count, uppercase and lowercase forms, first and last word, search count and a replaced version of the feedback.

Requirements

- Write and execute the Python program.
- Display the requested information as a clear dashboard.
- Test the program with different input values.

Algorithm

1. Read/prepare the required data.

2. Process the data using the prescribed Python concept.

3. Calculate the required results.

4. Display the dashboard and verify the result.

Python Program

```
feedback=input("Feedback: "); search=input("Word to search: "); repl=input("Replacement: ")
words=feedback.split()
print("\n========================================")
print("      UNIVERSITY FEEDBACK DASHBOARD")
print("========================================")
print("Characters:",len(feedback)); print("Words:",len(words))
print("Uppercase:",feedback.upper()); print("Lowercase:",feedback.lower())
print("First Word:",words[0] if words else ""); print("Last Word:",words[-1] if words else "")
print("Search Count:",feedback.lower().count(search.lower()))
print("Updated:",feedback.replace(search,repl))
print("========================================")
```

Sample Dashboard Output

```
========================================
      UNIVERSITY FEEDBACK DASHBOARD
========================================
Characters: 33
Words: 5
Uppercase: PYTHON LABORATORY IS VERY USEFUL
Lowercase: python laboratory is very useful
First Word: Python
Last Word: useful
Search Count: 1
Updated: Programming laboratory is very useful
========================================
```

Result

The program was executed successfully and the required dashboard was generated.

Viva / Practice Questions

1. What is string slicing?

2. What does split() return?

3. Difference between find() and count()?

Experiment 6: Airport Passenger Management Dashboard

Syllabus Topic: Lists: accessing, operations and methods

Question / Problem Statement

Write a Python program to manage an airport passenger list. Provide options to add, remove, search, sort and display passengers and show the total passenger count.

Requirements

- Write and execute the Python program.
- Display the requested information as a clear dashboard.
- Test the program with different input values.

Algorithm

1. Read/prepare the required data.

2. Process the data using the prescribed Python concept.

3. Calculate the required results.

4. Display the dashboard and verify the result.

Python Program

```
passengers=["Ayesha","Rahul","Fatima","Sameer"]
while True:
    print("\n1.Add  2.Remove  3.Search  4.Sort  5.Dashboard  6.Exit")
    ch=input("Choice: ")
    if ch=="1": passengers.append(input("Name: "))
    elif ch=="2":
        name=input("Name: ")
        if name in passengers: passengers.remove(name)
        else: print("Not found")
    elif ch=="3":
        name=input("Name: "); print("FOUND" if name in passengers else "NOT FOUND")
    elif ch=="4": passengers.sort()
    elif ch=="5":
        print("\n====================================")
        print("      AIRPORT PASSENGER DASHBOARD")
        print("====================================")
        print("Total Passengers:",len(passengers))
        for i,p in enumerate(passengers,1): print(i,".",p)
    elif ch=="6": break
    else: print("Invalid choice")
```

Sample Dashboard Output

```
====================================
      AIRPORT PASSENGER DASHBOARD
====================================
Total Passengers: 4
1 . Ayesha
2 . Fatima
3 . Rahul
4 . Sameer
====================================
```

Result

The program was executed successfully and the required dashboard was generated.

Viva / Practice Questions

1. What is a list?

2. How does append() work?

3. How can a passenger be searched?

4. What does sort() do?

Experiment 7: Restaurant Order Management Dashboard

Syllabus Topic: Functions: defining, calling, arguments and return values

Question / Problem Statement

Write a Python program using functions to manage a restaurant order. Create functions for bill calculation, tax calculation and dashboard display.

Requirements

- Write and execute the Python program.
- Display the requested information as a clear dashboard.
- Test the program with different input values.

Algorithm

1. Read/prepare the required data.

2. Process the data using the prescribed Python concept.

3. Calculate the required results.

4. Display the dashboard and verify the result.

Python Program

```
def item_amount(price,qty): return price*qty
def tax(amount,rate=5): return amount*rate/100
def dashboard(table,subtotal,tax_amount):
    print("\n====================================")
    print("       RESTAURANT ORDER DASHBOARD")
    print("====================================")
    print("Table:",table); print("Subtotal: ₹",subtotal)
    print("Tax: ₹",tax_amount); print("Total: ₹",subtotal+tax_amount)
menu={"Biryani":250,"Juice":80,"Dessert":120,"Coffee":60}
table=input("Table: "); subtotal=0
for item,price in menu.items():
    qty=int(input(item+" quantity: "))
    subtotal += item_amount(price,qty)
tax_amount=tax(subtotal)
dashboard(table,subtotal,tax_amount)
```

Sample Dashboard Output

```
====================================
       RESTAURANT ORDER DASHBOARD
====================================
Table: 12
Subtotal: ₹ 780
Tax: ₹ 39.0
Total: ₹ 819.0
====================================
```

Result

The program was executed successfully and the required dashboard was generated.

Viva / Practice Questions

1. What is a function?

2. What is a return value?

3. What is a default argument?

Experiment 8: Job Mela Candidate Selection Dashboard

Syllabus Topic: Function arguments, anonymous/lambda functions, local/global variables

Question / Problem Statement

Write a Python program to maintain Job Mela candidates. Define a function to check eligibility and use a lambda expression to sort eligible candidates by percentage in descending order.

Requirements

- Write and execute the Python program.
- Display the requested information as a clear dashboard.
- Test the program with different input values.

Algorithm

1. Read/prepare the required data.

2. Process the data using the prescribed Python concept.

3. Calculate the required results.

4. Display the dashboard and verify the result.

Python Program

```
def eligible(p): return p>=60
candidates=[("Ahmed",86),("Sara",72),("Rahul",58),("Ayesha",91),("Sameer",67)]
selected=[{"name":n,"percentage":p} for n,p in candidates if eligible(p)]
selected=sorted(selected,key=lambda x:x["percentage"],reverse=True)
print("\n========================================")
print("          JOB MELA DASHBOARD")
print("========================================")
print("Total Candidates:",len(candidates))
print("Eligible:",len(selected))
for i,c in enumerate(selected,1):
    print(i,c["name"],"-",c["percentage"],"%")
print("========================================")
```

Sample Dashboard Output

```
========================================
          JOB MELA DASHBOARD
========================================
Total Candidates: 5
Eligible: 4
1 Ayesha - 91 %
2 Ahmed - 86 %
3 Sara - 72 %
4 Sameer - 67 %
========================================
```

Result

The program was executed successfully and the required dashboard was generated.

Viva / Practice Questions

1. What is a lambda function?

2. Why is lambda useful in sorted()?

3. What is a local variable?

Experiment 9: University Administration Dashboard – Modules & Packages

Syllabus Topic: Functions/modules, importing modules, packages and composition

Question / Problem Statement

Create a small University Administration application using separate modules for student, admission and placement information. Import those modules into a main program and display a combined dashboard.

Requirements

- Write and execute the Python program.
- Display the requested information as a clear dashboard.
- Test the program with different input values.

Algorithm

1. Read/prepare the required data.

2. Process the data using the prescribed Python concept.

3. Calculate the required results.

4. Display the dashboard and verify the result.

Python Program

```
# student.py
def student_count(): return 4850

# admission.py
def admission_count(): return 1250

# placement.py
def placed_count(): return 620

# main.py
from student import student_count
from admission import admission_count
from placement import placed_count

print("\n====================================")
print("      UNIVERSITY ADMIN DASHBOARD")
print("====================================")
print("Students :",student_count())
print("Admissions:",admission_count())
print("Placed    :",placed_count())
print("====================================")
```

Sample Dashboard Output

```
====================================
      UNIVERSITY ADMIN DASHBOARD
====================================
Students : 4850
Admissions: 1250
Placed    : 620
====================================
```

Result

The program was executed successfully and the required dashboard was generated.

Viva / Practice Questions

1. What is a module?

2. How is a function imported from a module?

3. What is a package?

Experiment 10: University Admission Validation Dashboard

Syllabus Topic: Regular expressions: match, search, modifiers and patterns

Question / Problem Statement

Write a Python program to validate a university admission applicant's email, mobile number and application ID using regular expressions. Display VALID/INVALID status and the final application status.

Requirements

- Write and execute the Python program.
- Display the requested information as a clear dashboard.
- Test the program with different input values.

Algorithm

1. Read/prepare the required data.

2. Process the data using the prescribed Python concept.

3. Calculate the required results.

4. Display the dashboard and verify the result.

Python Program

```
import re
name=input("Applicant: "); email=input("Email: ")
mobile=input("Mobile: "); app=input("Application ID (APP1234): ")
email_ok=bool(re.match(r'^[\w.-]+@[\w.-]+\.\w+$',email))
mobile_ok=bool(re.match(r'^[6-9]\d{9}$',mobile))
app_ok=bool(re.match(r'^APP\d{4}$',app))
status="ACCEPTED" if email_ok and mobile_ok and app_ok else "RECHECK DETAILS"
print("\n====================================")
print("      UNIVERSITY ADMISSION DASHBOARD")
print("====================================")
print("Applicant:",name)
print("Email:", "VALID" if email_ok else "INVALID")
print("Mobile:", "VALID" if mobile_ok else "INVALID")
print("Application ID:", "VALID" if app_ok else "INVALID")
print("Application:",status)
print("====================================")
```

Sample Dashboard Output

```
====================================
      UNIVERSITY ADMISSION DASHBOARD
====================================
Applicant: Ahmed
Email: VALID
Mobile: VALID
Application ID: VALID
Application: ACCEPTED
====================================
```

Result

The program was executed successfully and the required dashboard was generated.

Viva / Practice Questions

1. What is a regular expression?

2. What does \d represent?

3. What is the difference between match() and search()?

Experiment 11: Festival & National Event File Dashboard

Syllabus Topic: File handling: opening, reading, writing and closing files

Question / Problem Statement

Write a Python program to register participants for a Festival/National Event. Store participant names and institutions in a file, read the file and display total registrations and participant details.

Requirements

- Write and execute the Python program.
- Display the requested information as a clear dashboard.
- Test the program with different input values.

Algorithm

1. Read/prepare the required data.

2. Process the data using the prescribed Python concept.

3. Calculate the required results.

4. Display the dashboard and verify the result.

Python Program

```
filename="event_participants.txt"
n=int(input("Number of participants: "))
with open(filename,"a") as f:
    for i in range(n):
        name=input("Name: "); institution=input("Institution: ")
        f.write(name+" | "+institution+"\n")
with open(filename,"r") as f:
    records=f.readlines()
print("\n========================================")
print("       NATIONAL EVENT DASHBOARD")
print("========================================")
print("Total Registrations:",len(records))
for i,r in enumerate(records,1): print(i,r.strip())
print("========================================")
```

Sample Dashboard Output

```
========================================
       NATIONAL EVENT DASHBOARD
========================================
Total Registrations: 3
1 Ayesha | MANUU
2 Rahul | MANUU
3 Sara | GRIET
========================================
```

Result

The program was executed successfully and the required dashboard was generated.

Viva / Practice Questions

1. Difference between r, w and a modes?

2. Why use with open()?

3. How can file records be counted?

Experiment 12: Seminar, Workshop & Hackathon File Dashboard

Syllabus Topic: Reading/writing files and data processing

Question / Problem Statement

Write a Python program to maintain Seminar, Workshop and Hackathon records in a file. Read the records and generate a dashboard showing event counts and total participants.

Requirements

- Write and execute the Python program.
- Display the requested information as a clear dashboard.
- Test the program with different input values.

Algorithm

1. Read/prepare the required data.

2. Process the data using the prescribed Python concept.

3. Calculate the required results.

4. Display the dashboard and verify the result.

Python Program

```
filename="academic_events.txt"
events=[("Seminar","AI & Data Science",120),
("Workshop","Python Programming",80),
("Hackathon","Innovation Challenge",150),
("Seminar","Machine Learning",100),
("Workshop","Cyber Security",75)]
with open(filename,"w") as f:
    for typ,title,p in events: f.write(f"{typ}|{title}|{p}\n")
counts={"Seminar":0,"Workshop":0,"Hackathon":0}; total=0
with open(filename,"r") as f:
    for line in f:
        typ,title,p=line.strip().split("|"); counts[typ]+=1; total+=int(p)
print("\n========================================")
print(" SEMINAR / WORKSHOP / HACKATHON DASHBOARD")
print("========================================")
print("Seminars:",counts["Seminar"])
print("Workshops:",counts["Workshop"])
print("Hackathons:",counts["Hackathon"])
print("Total Participants:",total)
print("========================================")
```

Sample Dashboard Output

```
========================================
 SEMINAR / WORKSHOP / HACKATHON DASHBOARD
========================================
Seminars: 2
Workshops: 2
Hackathons: 1
Total Participants: 525
========================================
```

Result

The program was executed successfully and the required dashboard was generated.

Viva / Practice Questions

1. Why are files useful for event records?

2. How is split() used while reading a record?

3. How would you add a new event category?

Experiment 13: University Database Management Dashboard

Syllabus Topic: Database introduction, connections and executing queries

Question / Problem Statement

Using Python sqlite3, create a University Student database. Create a student table, insert records, retrieve records using SQL queries and display total students, average CGPA and topper information.

Requirements

- Write and execute the Python program.
- Display the requested information as a clear dashboard.
- Test the program with different input values.

Algorithm

1. Read/prepare the required data.

2. Process the data using the prescribed Python concept.

3. Calculate the required results.

4. Display the dashboard and verify the result.

Python Program

```
import sqlite3
conn=sqlite3.connect("university.db"); cur=conn.cursor()
cur.execute("""CREATE TABLE IF NOT EXISTS student(
student_id INTEGER PRIMARY KEY,name TEXT,department TEXT,cgpa REAL)""")
data=[(101,"Ayesha","CSE",8.8),(102,"Rahul","IT",7.9),
(103,"Sara","CSE",9.1),(104,"Sameer","ECE",8.2)]
cur.executemany("INSERT OR IGNORE INTO student VALUES(?,?,?,?)",data)
conn.commit()
cur.execute("SELECT COUNT(*) FROM student"); total=cur.fetchone()[0]
cur.execute("SELECT AVG(cgpa) FROM student"); avg=cur.fetchone()[0]
cur.execute("SELECT name,cgpa FROM student ORDER BY cgpa DESC LIMIT 1"); top=cur.fetchone()
print("\n========================================")
print("      UNIVERSITY DATABASE DASHBOARD")
print("========================================")
print("Total Students:",total); print("Average CGPA:",round(avg,2))
print("Top Student:",top[0]); print("Top CGPA:",top[1])
print("========================================")
conn.close()
```

Sample Dashboard Output

```
========================================
      UNIVERSITY DATABASE DASHBOARD
========================================
Total Students: 4
Average CGPA: 8.5
Top Student: Sara
Top CGPA: 9.1
========================================
```

Result

The program was executed successfully and the required dashboard was generated.

Viva / Practice Questions

1. What is a database connection?

2. What does execute() do?

3. Why is commit() required after insertion?

Experiment 14: Placement, Training & Alumni Database Dashboard

Syllabus Topic: Database queries, transactions and error handling

Question / Problem Statement

Develop an integrated Python application for Placement, Training and Alumni management. Use SQLite tables, SQL queries, transactions and exception handling. Insert records and display placement, training and alumni statistics.

Requirements

- Write and execute the Python program.
- Display the requested information as a clear dashboard.
- Test the program with different input values.

Algorithm

1. Read/prepare the required data.

2. Process the data using the prescribed Python concept.

3. Calculate the required results.

4. Display the dashboard and verify the result.

Python Program

```
import sqlite3
try:
    conn=sqlite3.connect("career_management.db"); cur=conn.cursor()
    cur.execute("CREATE TABLE IF NOT EXISTS placement(id INTEGER PRIMARY KEY,company TEXT,package
REAL,status TEXT)")
    cur.execute("CREATE TABLE IF NOT EXISTS training(id INTEGER PRIMARY KEY,name TEXT,participants
INTEGER)")
    cur.execute("CREATE TABLE IF NOT EXISTS alumni(id INTEGER PRIMARY KEY,name TEXT,company TEXT)")
    cur.execute("INSERT OR IGNORE INTO placement VALUES(1,'ABC Technologies',8.5,'Placed')")
    cur.execute("INSERT OR IGNORE INTO training VALUES(1,'Python & AI Training',120)")
    cur.execute("INSERT OR IGNORE INTO alumni VALUES(1,'Rahul','XYZ Ltd')")
    conn.commit()
    cur.execute("SELECT COUNT(*) FROM placement WHERE status='Placed'"); placed=cur.fetchone()[0]
    cur.execute("SELECT SUM(participants) FROM training"); trained=cur.fetchone()[0] or 0
    cur.execute("SELECT COUNT(*) FROM alumni"); alumni=cur.fetchone()[0]
    cur.execute("SELECT MAX(package) FROM placement"); highest=cur.fetchone()[0] or 0
    print("\n================================================")
    print("       PLACEMENT, TRAINING & ALUMNI")
    print("                  DASHBOARD")
    print("================================================")
    print("Students Placed:",placed); print("Students Trained:",trained)
    print("Registered Alumni:",alumni); print("Highest Package: ₹",highest,"LPA")
    print("================================================")
except sqlite3.Error as e:
    if "conn" in locals(): conn.rollback()
    print("Database Error:",e)
finally:
    if "conn" in locals(): conn.close()
```

Sample Dashboard Output

```
================================================
       PLACEMENT, TRAINING & ALUMNI
                  DASHBOARD
================================================
Students Placed: 1
Students Trained: 120
Registered Alumni: 1
```

```
Highest Package: ₹ 8.5 LPA
================================================
```

Result

The program was executed successfully and the required dashboard was generated.

Viva / Practice Questions

1. Why are transactions used?

2. What does rollback() do?

3. Why is try-except important in database programs?

4. How could an Uber ride table be added to this application?

Laboratory Submission Checklist

- Write the Question/Problem Statement in the laboratory record.
- Understand the algorithm before execution.
- Run the program with different input values.
- Record the dashboard output.
- Answer the viva questions.
- Keep generated files/database files for Experiments 11–14 with the laboratory work.

End of 14-Experiment Python Programming Laboratory Manual